Desktop AI-Powered Payroll Management System with Salary Prediction

Abstract

This project develops a desktop payroll management system integrated with AI-based salary prediction. It automates payroll, deductions, tax calculations, attendance integration, and reporting while using machine learning to estimate future salaries.

Objectives

Automate payroll; manage employee records; generate salary slips; predict future salaries using AI; improve accuracy and reduce processing time.

Modules

Employee Management, Attendance, Payroll Processing, Tax & Deductions, AI Salary Prediction, Reports, User Authentication.

Data Preprocessing

Collected payroll data, removed missing values, encoded categorical variables, normalized numerical features, and split data into training and testing datasets.

AI Model

A regression model is trained on employee features such as experience, performance, designation, attendance, and previous salary to predict future salary.

Tools

Python, Tkinter, Pandas, NumPy, Scikit-learn, SQLite/MySQL, Matplotlib.

Testing

Unit, integration, functional, and user acceptance testing confirmed correct payroll calculations and prediction accuracy.

Results

The system successfully generates payroll, calculates deductions, produces payslips, and predicts salaries with improved efficiency.

Conclusion

The project demonstrates that AI can enhance payroll management by automating routine tasks and supporting salary planning decisions.

Future Scope

Cloud deployment, biometric integration, deep learning models, mobile app, and HR analytics dashboard.